

Lab#6 (Op-amp as a Low Pass & High Pass Active filter and p-n Diode)

- EXPERIMENT #1: Low Pass Active Filter
- EXPERIMENT #2: High Pass Active Filter
- EXPERIMENT #3: I(V) characteristics of a p-n diode
- EXPERIMENT #4: I(V) Determine the response of a circuit with a p-n diode

Experiment #1: Low Pass Active Filter

Construct the following circuit, using +/-15 V power supplies for the Op-amp.

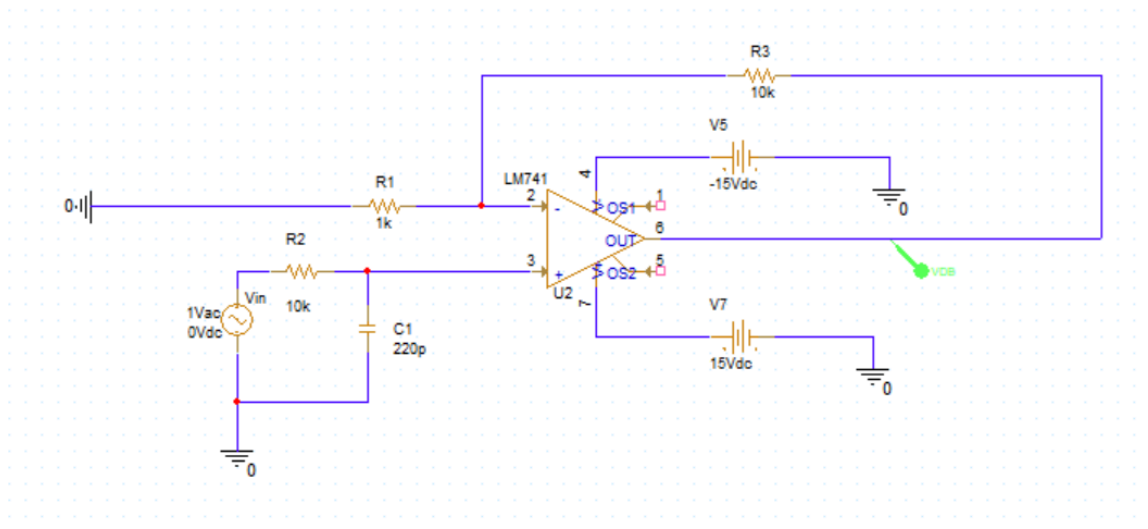
1. Measure and record the actual values of the components using an ohmmeter and capacitance meter.

R1 (actual) = 0.98595KOHM

R2 (actual) = 9.8633KOHM

R3(actual) = 9.8539KOHM

C (actual) = 208.1pF



2. Build the circuit in PSpice and find your cut-off frequency.
3. Using the oscilloscope, measure both the input and the output voltages (magnitude) as a function of frequency for the different frequencies shown in the Table below. Draw the graph and compare your cutoff frequency and voltage gain(dB) with the PSPice graph and hand calculation.

f (measured) = 14.3157Hz

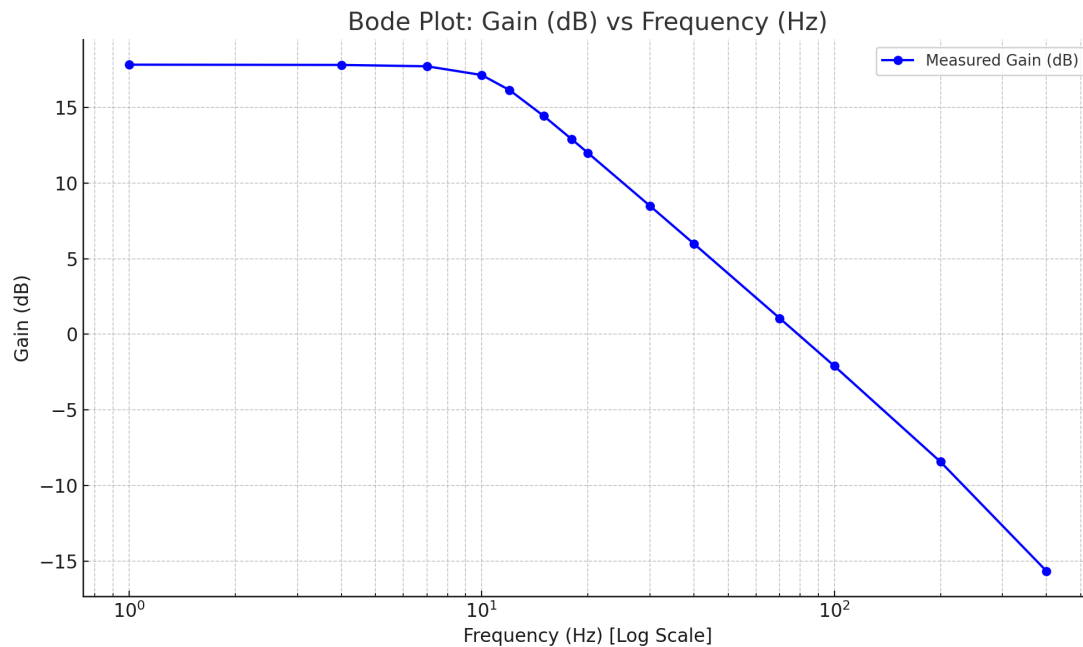
f (calculated) = 77540Hz

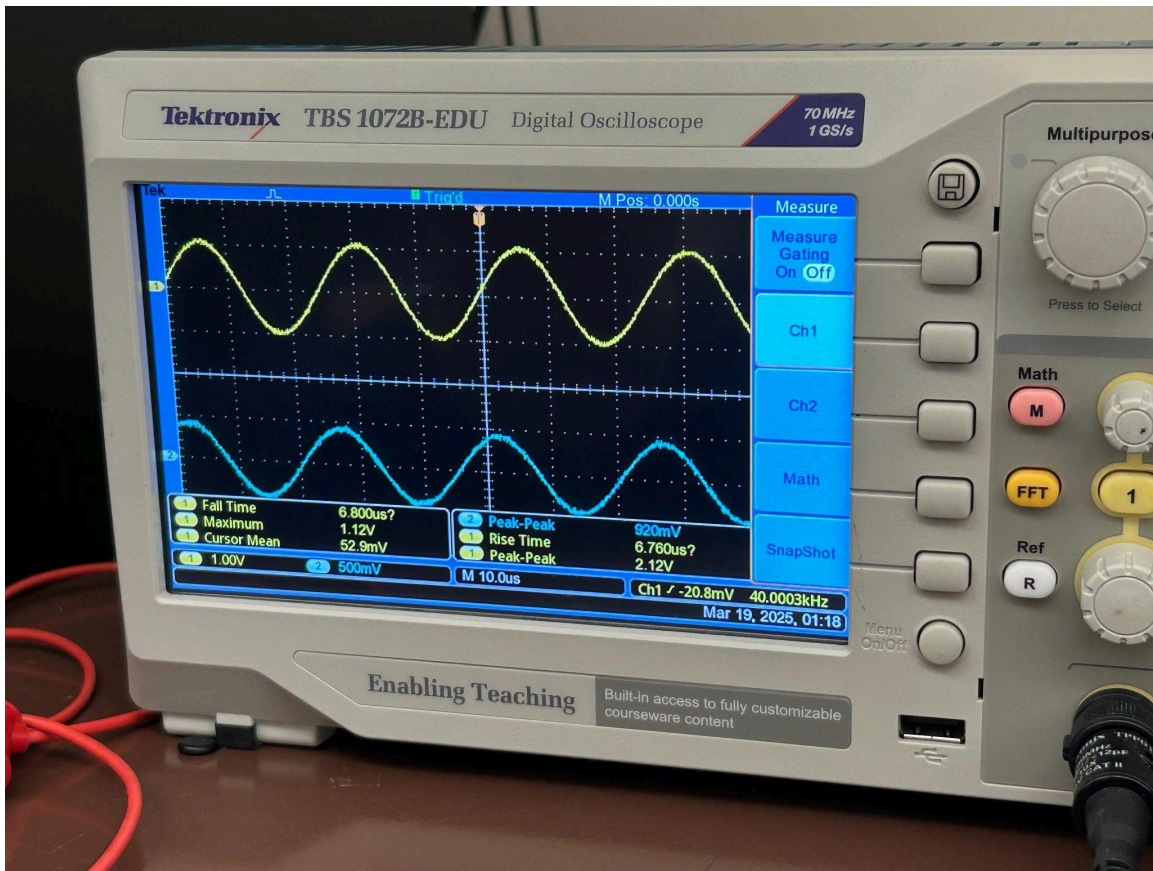
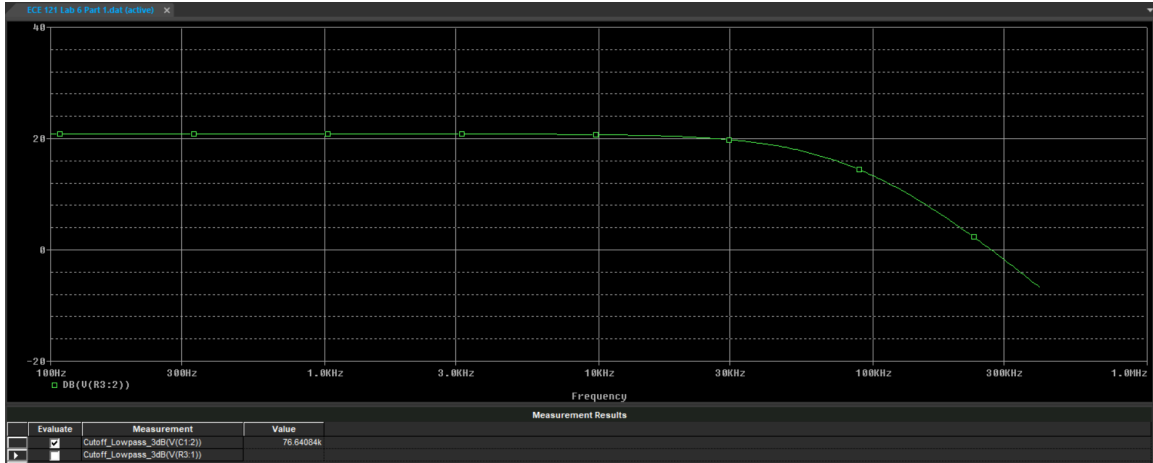
f (simulated) = 76604Hz

Gain(dB) (measured) = 14.8334dB

Gain(dB) (calculated) = 14.7246dB

Gain(dB) (simulated) = 14.7434dB





1.) Measured

Highest value: 17.8374 dB
at 1 Hz

-3dB cutoff = 17.8374
= 14.8334 dB

$G_{\text{ain}}(\text{dB}) = 14.8334 \text{ dB}$

$f_c = 12 + \frac{14.83 - 16.15}{14.44 - 16.15} \cdot (15 - 12)$

$f_c = 14.3157 \text{ Hz}$

Calculated

$G_{\text{ain}}(\text{dB}) = 20 \log \left(1 + \frac{9.85394}{0.98594} \right) \cdot \frac{1}{\sqrt{2}}$

$G_{\text{ain}}(\text{dB}) = 14.7246 \text{ dB}$

$f_c = \frac{1}{2\pi R_2 C}$

$f_c = \frac{1}{2\pi \cdot 9.86324 \text{ ohm} \cdot 208.1 \text{ pF}}$

$f_c = 77540 \text{ Hz}$

Suggested frequency (kHz)	Actual frequency	Vs	Vo	Vo/ Vs	(Vo/ Vs) dB
1	1KHz	1V	7.7924V	7.7924V	17.83342
4	4 KHz	1V	7.777V	7.777V	17.81624
7	7 KHz	1V	7.696V	7.696V	17.7253
10	10 KHz	1V	7.2024V	7.2024V	17.14954
12	12 KHz	1V	6.4177V	6.4177V	16.14759
15	15 KHz	1V	5.2693V	5.2693V	14.43506
18	18 KHz	1V	4.4174V	4.4174V	12.90333
20	20 KHz	1V	3.9810V	3.981V	11.99984
30	30 KHz	1V	2.6587V	2.6587V	8.493387

40	40 KHz	1V	1.9913V	1.9913V	5.982734
70	70 KHz	1V	1.1297V	1.1297V	1.059263
100	100 KHz	1V	0.78523V	0.78523V	-2.10006
200	200 KHz	1V	0.3786V	0.3786V	-8.43639
400	400 KHz	1V	0.1645V	0.1645V	-15.6767

Conclusion: I think that this experiment went well however there were some inconsistencies with my measured frequencies and calculated/simulated frequencies. This could have been human/equipment error or some unknown characteristics from the operation amplifier.

Experiment #2: High Pass Active Filter

Construct the following circuit, using +/-15 V power supplies for the Op-amp.

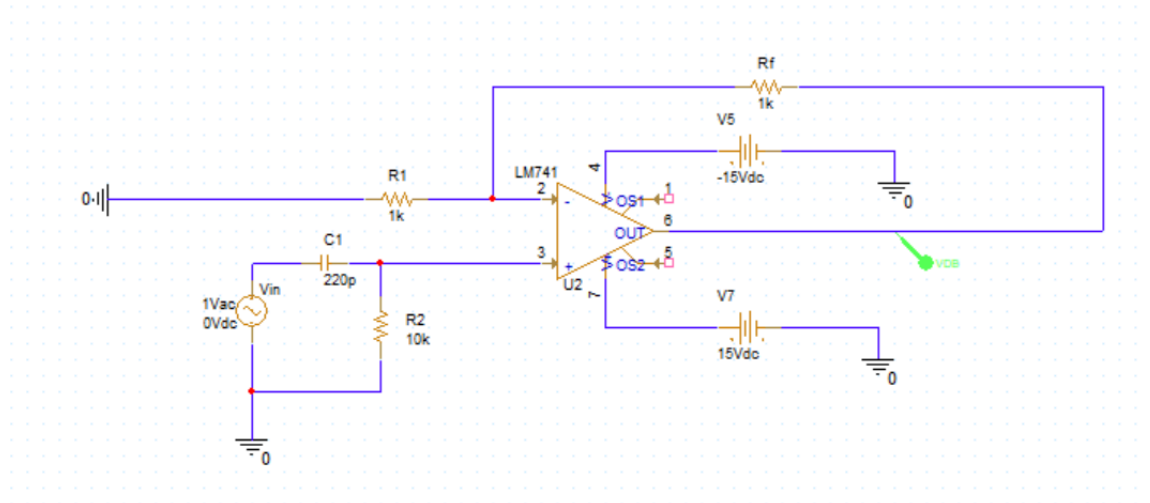
1. Measure and record the actual values of the components using an ohmmeter and capacitance meter.

R1 (actual) = 0.98595KOHM

R2 (actual) = 9.8633KOHM

RF (actual) = 0.98521KOHM

C (actual) = 208.1pF



2. Build the circuit in PSpice and find your cut-off frequency.
3. Using the oscilloscope, measure both the input and the output voltages (magnitude) as a function of frequency for the different frequencies shown in the Table below. Draw the graph and compare your cutoff frequency and voltage gain(dB) with the PSpice graph and hand calculation.

$$f(\text{measured}) = \underline{94.55\text{Hz}}$$

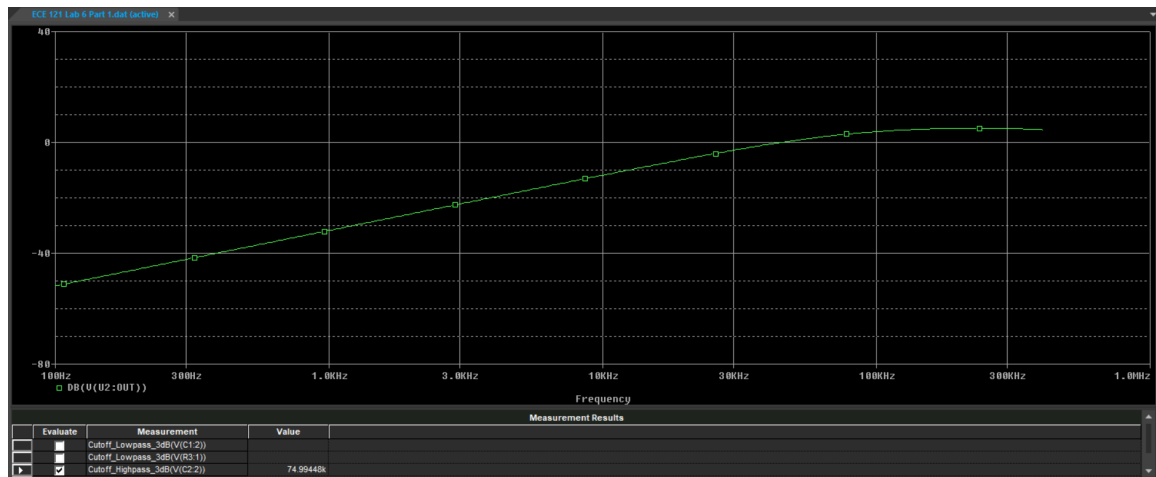
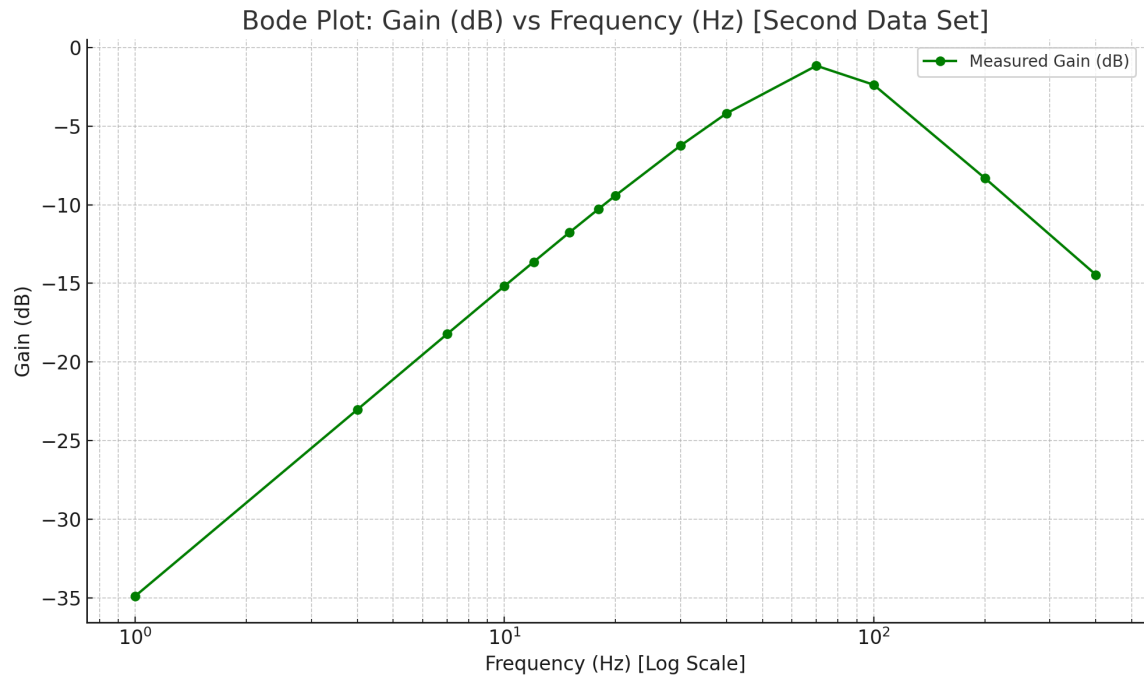
$$f(\text{calculated}) = \underline{77540\text{Hz}}$$

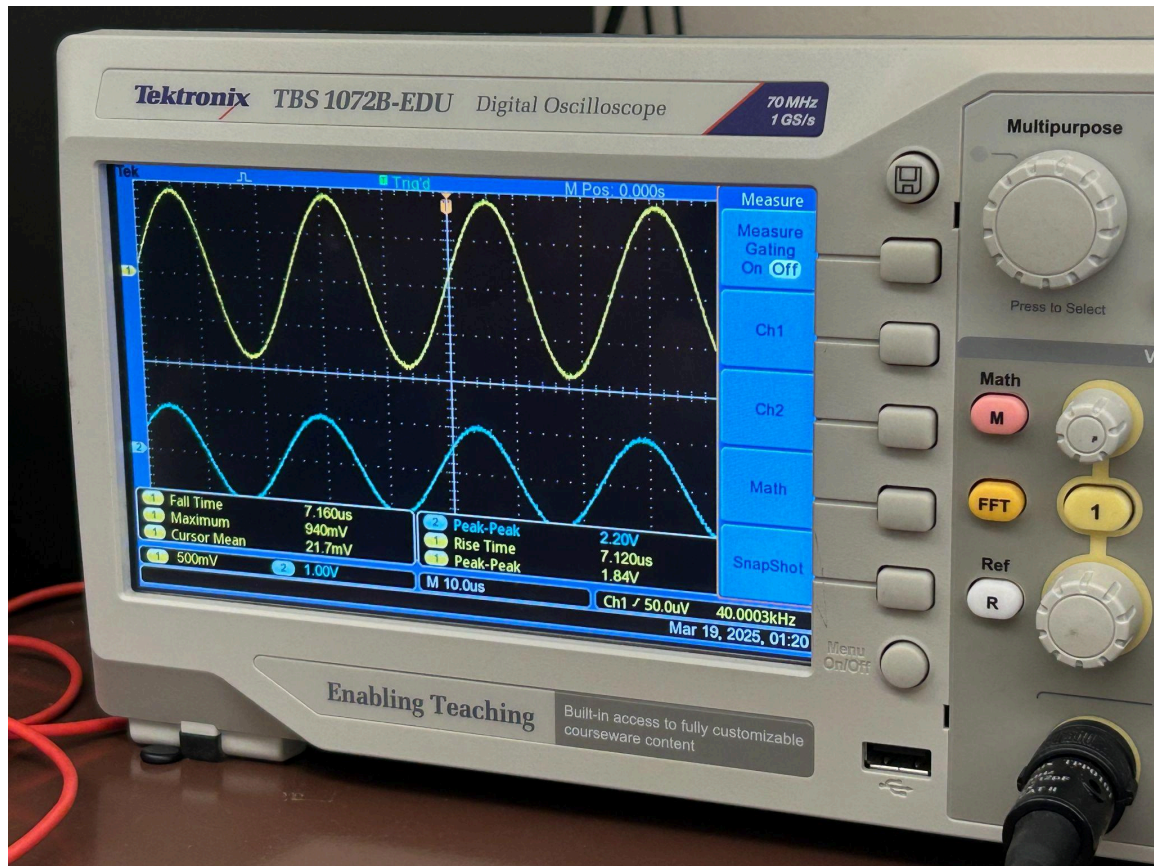
$$f(\text{simulated}) = \underline{74994\text{Hz}}$$

$$\text{Gain(dB)}(\text{measured}) = \underline{-4.1537857\text{dB}}$$

$$\text{Gain(dB)}(\text{calculated}) = \underline{4.2549\text{dB}}$$

$$\text{Gain(dB)}(\text{simulated}) = \underline{4.23549\text{dB}}$$





2.) Measured

Highest Value: -1.1537857

$\text{Gain (dB)} = -4.1537857$

$f_c = 100 + \frac{-4.1538 - (-2.3626)}{-8.2985 - (-2.3626)} \cdot (200 - 100)$

$f_c = 100 + \frac{-1.7912}{-5.9359} \cdot 100$

$f_c = 94.55 \text{ Hz}$

Calculated

$\text{Gain (dB)} = 20 \log \left(1 + \frac{0.985214}{0.985456} \right) \cdot \frac{1}{\sqrt{2}}$

$\text{Gain (dB)} = 4.2549$

$f_c = \frac{1}{2\pi R_2 C}$

$f_c = \frac{1}{2\pi \cdot 9.8633 \mu\text{s} \cdot 208.1 \text{ pF}}$

$f_c = 77540 \text{ Hz}$

Suggested frequency (kHz)	Actual frequency	Vs	Vo	Vo/ Vs	(Vo/ Vs) dB
1	1KHz	1V	17.991mV	0.017991	-34.898894

4	4 KHz	1V	70.591mV	0.070591	-23.025013
7	7 KHz	1V	0.12273V	0.12273	-18.220985
10	10 KHz	1V	0.17428V	0.17428	-15.175049
12	12 KHz	1V	0.208V	0.208	-13.638733
15	15 KHz	1V	0.25813V	0.25813	-11.76323
18	18 KHz	1V	0.30672V	0.30672	-10.265158
20	20 KHz	1V	0.33856V	0.33856	-9.4072871
30	30 KHz	1V	0.48766V	0.48766	-6.2376573
40	40 KHz	1V	0.61805V	0.61805	-4.1795278
70	70 KHz	1V	0.87561V	0.87561	-1.1537857
100	100 KHz	1V	0.76185V	0.76185	-2.3626106
200	200 KHz	1V	0.38466V	0.38466	-8.2984595
400	400 KHz	1V	0.18940V	0.1894	-14.452401

Conclusion: I think that this experiment went well however there were some inconsistencies with my measured frequencies and calculated/simulated frequencies. Also my measured dB cutoff is negative but the number correlates with the calculated/simulated dB cutoffs. This could have been human/equipment error or some unknown characteristics with the operation amplifier.

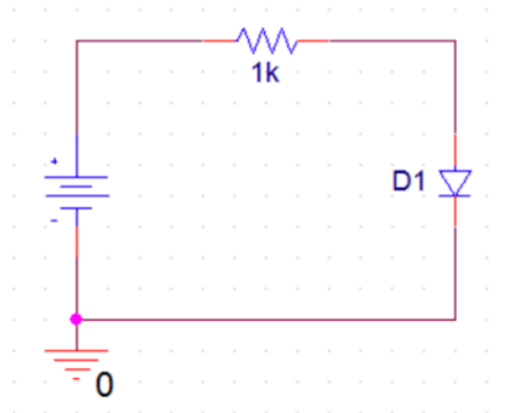
Experiment #3: I(V) characteristics of a p-n diode

R1=0.98553KOHM

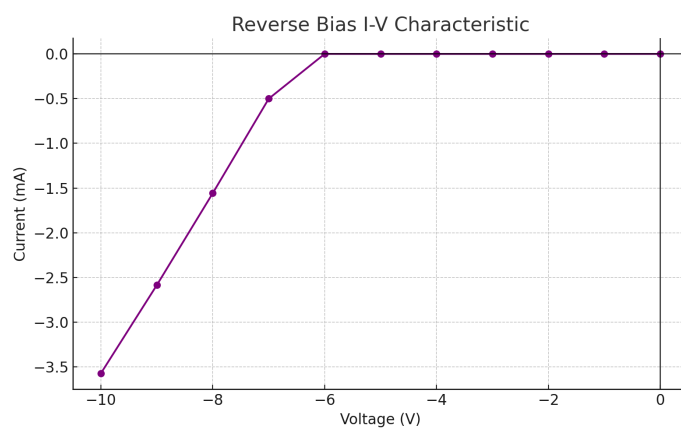
D1=1n5921B

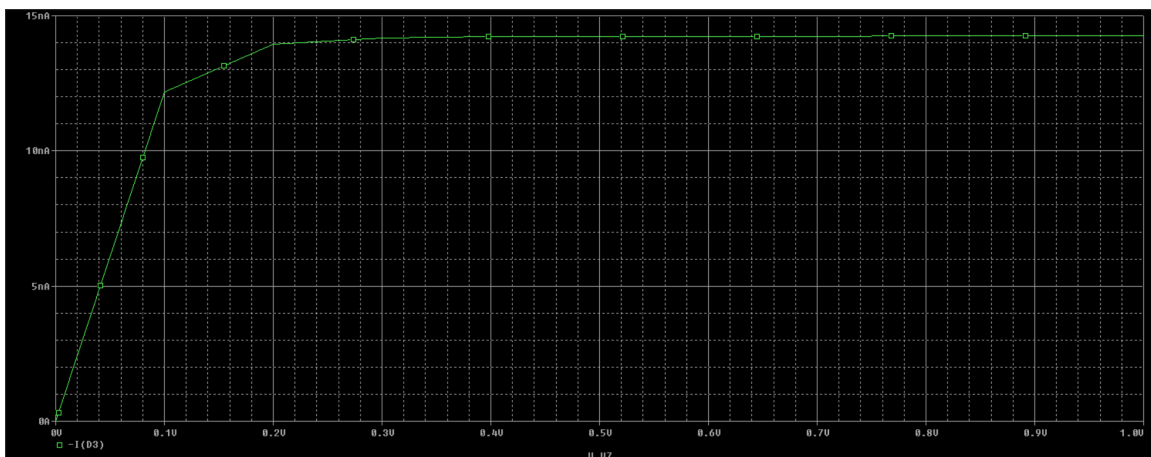
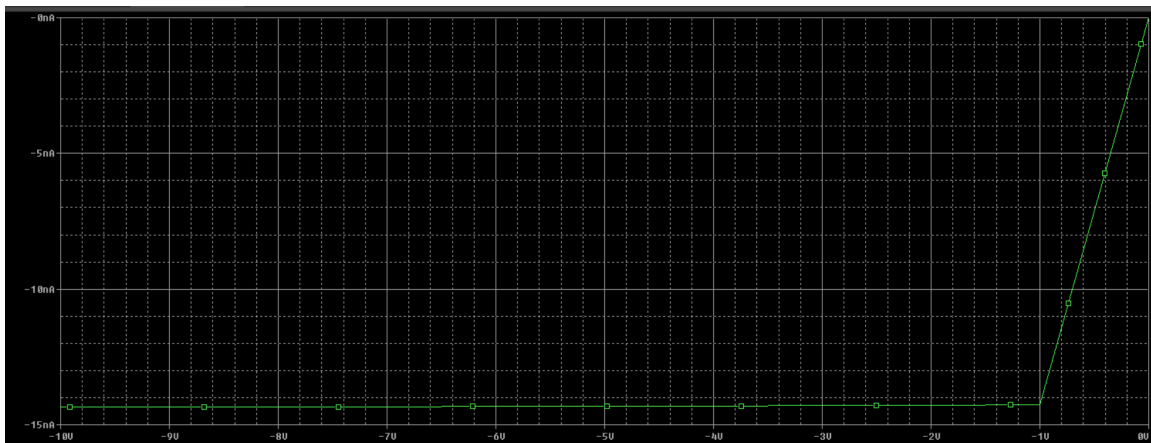
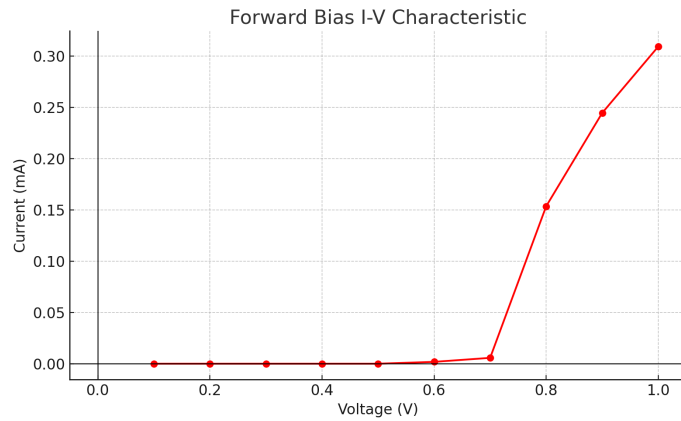
Set-up the measuring circuit shown below using a p-n diode:

- a) REVERSE region: vary the voltage in the range of -10 V to 0V in increments of 1 V. Measure the current for each voltage setting.
- b) FORWARD region: vary the voltage in the range of 0 V to 1V in increments of 0.1 V. Measure the current for each voltage setting.



Expierment #3	
a.)	
Voltage (V)	Current Reverse Bias
-10	-3.5718mA
-9	-2.5848mA
-8	-1.5568mA
-7	-0.5007mA
-6	0.0003mA
-5	0A
-4	0A
-3	0A
-2	0A
-1	0A
0	0A
b.)	Current Forward Bias
0.1	0A
0.2	0A
0.3	0A
0.4	0A
0.5	0A
0.6	.0018mA
0.7	0.0057mA
0.8	0.1535mA
0.9	0.2445mA
1	0.3092mA





Conclusion: I believe that this experiment went well because my measured graphs and simulated graphs match each other. This shows the ideal characteristics between a diode in forward and reverse bias.

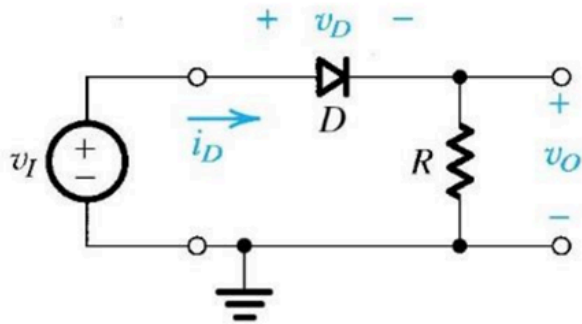
➤ **Experiment #4: Determine the response of a circuit with a p-n diode**

R1=0.98553KOHM

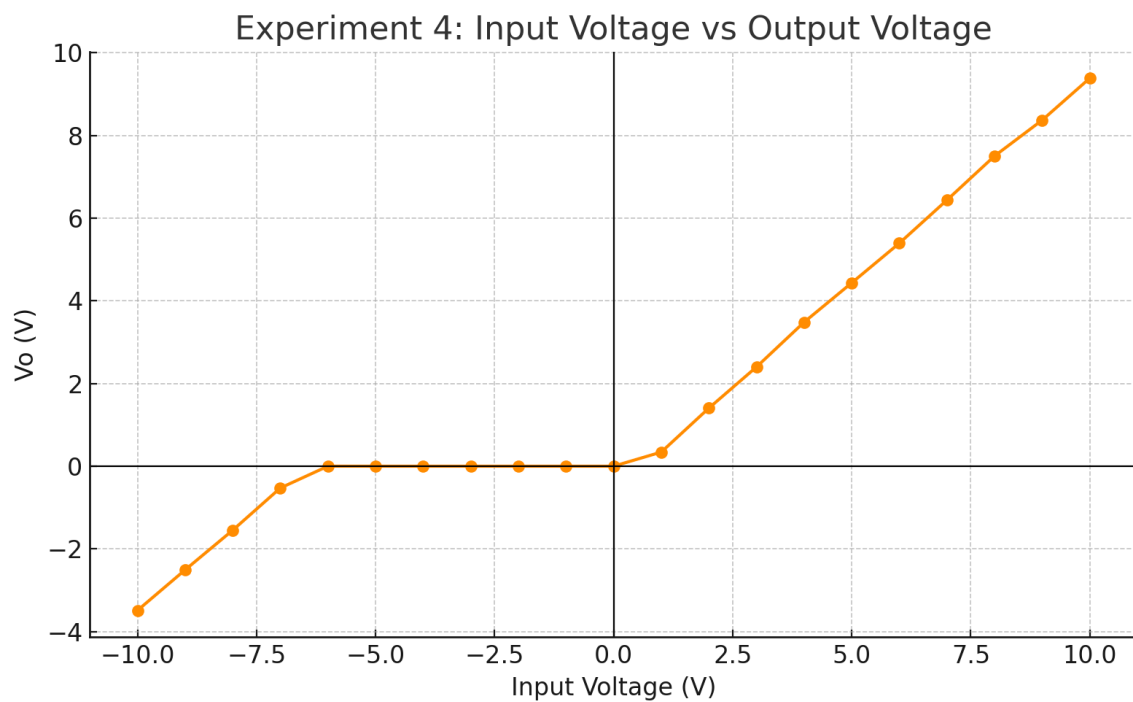
D1=1n5921B

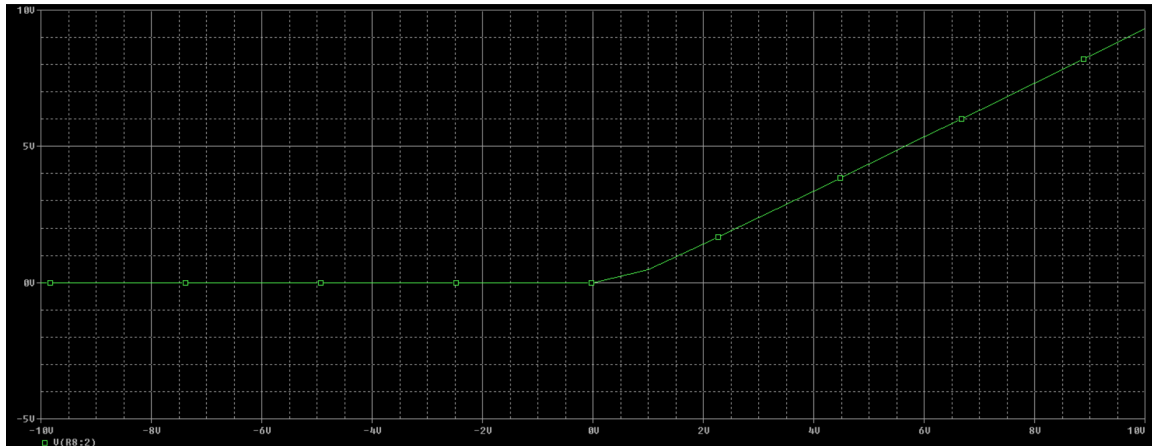
Set-up the measuring circuit shown below using a p-n diode:

Vary the voltage source from -10 V to + 10 V in 1 V increment and measure and record the output voltage.



Expierment 4	
Voltage (V)	Vo (V)
-10	-3.4884V
-9	-2.5085V
-8	-1.549V
-7	-0.52789V
-6	0.339mV
-5	0.020mV
-4	0.003mV
-3	0.001mV
-2	0V
-1	0V
0	0V
1	0.3407V
2	1.4039V
3	2.3993V
4	3.482V
5	4.4329V
6	5.392V
7	6.435V
8	7.5007V
9	8.3684V
10	9.3829V





Conclusion: I believe that this experiment went well because my measured graphs and simulated graphs match each other. This shows the response of the circuit with a voltage source ranging from -10V to 10V showing the relationship between negative source voltage and voltage output.